HW2: Generative Adversarial Network (GAN)

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Abstract

I train a DCGAN-style generative adversarial network (GAN) in PyTorch on the Kaggle WikiArt dataset (81,444 paintings at 128×128) and report hyperparameters, training curves, Inception Score (IS) and Fréchet Inception Distance (FID) from PyTorch-Ignite, and samples. The final model (hinge loss, spectral normalization, DiffAugment, 600 epochs) reaches an FID of 39.08.

1 Method

Model. A generator G maps $z \sim \mathcal{N}(0, I)$ to an image and a discriminator D scores images. In alternating steps I minimize the hinge losses

$$\mathcal{L}_D = \mathbb{E}_x[\max(0, 1 - D(T(x)))] + \mathbb{E}_z[\max(0, 1 + D(T(G(z))))], \qquad \mathcal{L}_G = -\mathbb{E}_z[D(T(G(z)))],$$

where T is DiffAugment (Zhao et al., 2020): random brightness, saturation, contrast, translation (up to $1/8$ of the image) and a half-size cutout, applied to real and generated images, with gradients flowing through T into G. I added it after the discriminator won too easily in an earlier BCE run (loss 0.002).

Architecture. G maps $z \in \mathbb{R}^{128}$ with a 4×4 transposed convolution to a 4^2 map and then five stride-2 4×4 transposed convolutions (512–256–128–64–32–3 channels, BatchNorm, ReLU, $\tanh$ output). D has five stride-2 4×4 convolutions (64–128–256–512–512, LeakyReLU 0.2) and a 4×4 convolution to one score, all with spectral normalization and no BatchNorm. The models have 3.8M and 7.0M parameters (Figure 1). I train with Adam, learning rate decayed linearly to zero over the second half, random horizontal flips, and an exponential moving average (decay 0.999) of the generator weights, which are used for all samples and metrics.

Evaluation. FID and IS use Inception-v3 (ImageNet weights) through PyTorch-Ignite, following the course blog; images are ImageNet-normalized and resized to 299×299. FID compares 5000

Preprint.

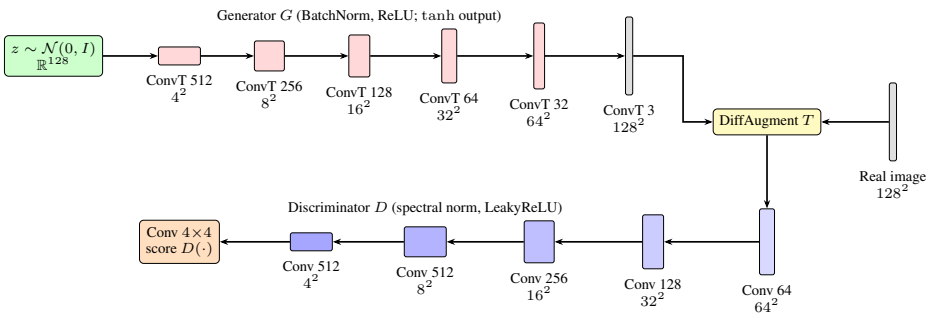


Figure 1: Architecture. Real and generated images pass through the same DiffAugment T before D. Box height is proportional to spatial size and width to the number of channels.

generated with 5000 random training images using the standard 2048-d pooled features (Ignite's default FID uses the 1000-d softmax output and has a much smaller scale, so I list it as secondary); IS uses the 5000 generated images.

2 Experiments

Data and settings. I use all 81,444 WikiArt images (labels unused), resized so the short side is 128 pixels, center-cropped to 128×128 and scaled to $[-1, 1]$. Table 1 lists the final settings (seed 0). I chose them with 14 runs (one seed each, 60 epochs unless stated) over loss (BCE or hinge), normalization (BatchNorm or spectral norm), DiffAugment, two-time-scale learning rates (TTUR), batch size, network width and training length. The 60-epoch FIDs ranged from 63 to 104, and the plain DCGAN (BCE, BatchNorm) reached 89.7.

Table 1: Hyperparameters and results of the final model.

Image size	128^2	Latent dim.	128	Batch size	64
Epochs	600 (763k it.)	Learning rate	2×10^{-4}	Adam β_1	0.5
Loss	hinge	D norm.	spectral	EMA decay	0.999
FID (↓)	**39.08**	IS, generated	4.84	IS, real images	8.54
FID, ignite default	0.0582	Final D loss	1.719	Final G loss	0.463

Training curves and samples. Figure 2 shows the training curves and Figure 3 real and generated images. The discriminator loss stays near its initial value of 2 (1.8 early, 1.72 at the end), so neither network overwhelms the other, and the FID on 2000 images falls from 125 at epoch 20 to 67 at epoch 120 and slowly to 55 at epoch 600 (the 5000-image FID is lower, 39.08).

3 Discussion

The samples have convincing palettes, brushwork and compositions (landscapes, skies, sketches), while faces and figures are often distorted and a few show small blocky artifacts. Inception-v3 fits paintings poorly (IS 4.84 for generated, 8.54 for real images), so I treat FID as the main metric.

Spectral norm with hinge loss and DiffAugment or batch size 64 gave FID 63–68 at 60 epochs, against about 90 for the plain DCGAN and most variants. DiffAugment helped less with BCE and BatchNorm (89.7 to 80.2) and not with BCE and spectral norm (91.5 to 93.4); TTUR (90.5) and a wider network (103.5) did not help. Training length paid off only with the better recipe: FID 63.1, 49.2 and 39.1 at 60, 150 and 600 epochs (the 60-epoch run had no learning-rate decay), while plain BCE at 150 epochs did not improve (90.8). Single runs are noisy: two seeds of one recipe gave 65.3 and 78.8, so gaps of about 10 points are not meaningful, and the final model is one seed.

Code. `https://github.com/Shubhanflash22/ECE-285-Deep-Generative-Models/tree/main/Homework/HW2`

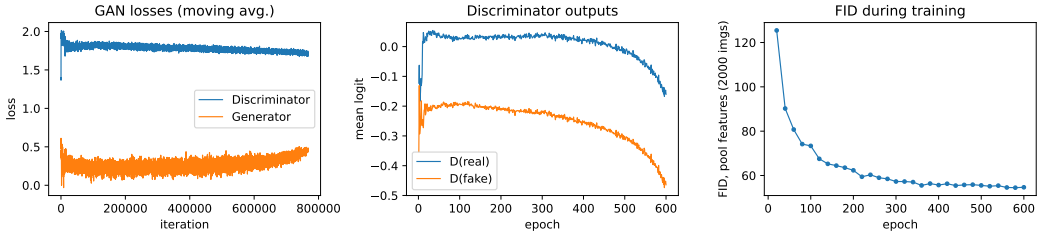


Figure 2: Training curves of the final run: GAN losses (moving average), discriminator scores and FID on 2000 images.

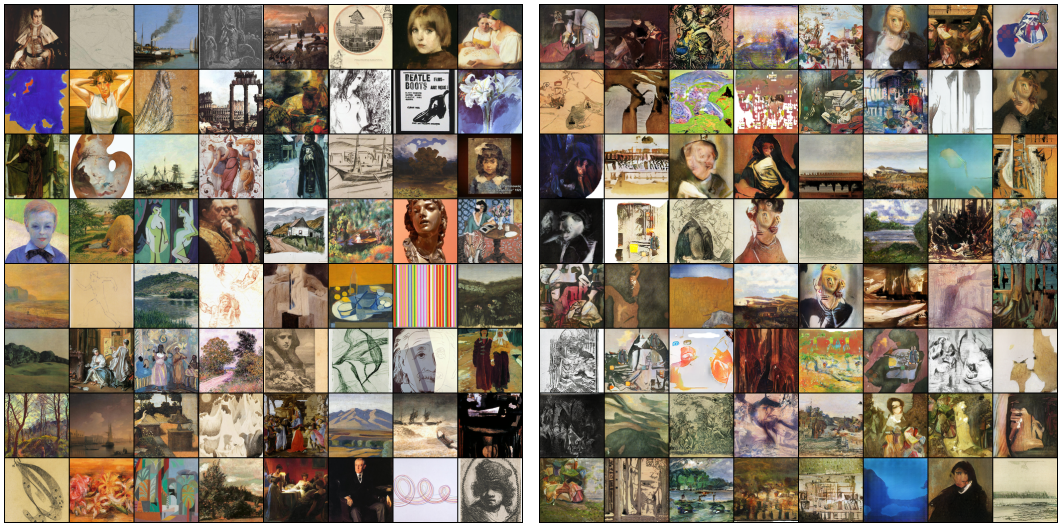


Figure 3: Real training images (left) and images generated by the final model (right).